

Yan Pan

📍 Sichuan, China ✉ yanpan@zohomail.com ☎ 13386384772 🔗 <https://marissapy.github.io/>

Key

- Founder and CEO of a synbio startup (TeloAetas Tech.)
- Founder and Leader of the UESTC BioMedical Student Club
- iGEM (International Genetically Engineered Machine competition) 2026 Ambassador of Asia-Pacific Region
- iGEM 2023 Competitor and 2024/2025 Team Advisor
- BGI Genomics Group Guest Young Scientist Program Teammates
- National Undergraduate Training Program on Innovation and Entrepreneurship, Team Leader
- Background in the intersection of computational biology(Systems Bio.) and synthetic biology
- Experienced in pre-market toxicology assessment for synbio-derived food ingredients in China

Education

BS **University of Electronic Science and Technology of China**, Clinical Medicine Sept 2022 – May 2027

- **GPA:** 3.8/4.0
- **Coursework:** Syetems Biology, Diagnostics, Surgery, Molecular Biology, Biochemistry, Cell Biology, Machine Learning Theory

Lab Experience

Center of Aging Reasarch, University of Electronic Science and Technology of China Chengdu, Sichuan
March 2023 – now

- Developing a virtual screening framework for anti-aging small molecules based on computational biology and systems biology.

Institute of Toxicology, Beijing Center for Disease Control and Prevention Beijing
Aug 2024 – Now

- Acquired *C. elegans*-based toxicity testing methods and contributed to the safety evaluation of synthetic biology-derived allulose prior to its market launch.

GPCR Pharmacology Lab, National Key Laboratory of Biotherapy, West China Hospital, Sichuan University Chengdu, Sichuan
Dec 2024 – Oct 2025

- Combined with structural biology, deep learning models are used to perform antibody design.

National Food Safety Risk Assessment Center, Chinese Academy of Medical Sciences and Peking Union Medical College Beijing/Remote
Aug 2024 – Jan 2025

- Conducted toxicity assessment of toxic substances that may exist in food, and made reference conclusions for food risk safety assessment.

Protein Engineering Lab, University of Electronic Science and Technology of China Chengdu, Sichuan
June 2022 – March 2023

- Using molecular simulation methods, the enzyme activity of PET hydrolase was modified and upgraded.

Publications

High-Throughput Screening for Ageing and Age-Related Disease Drug Discovery: Advances and Challenges *Ageing Research Reviews*
April 2026

Xingkun Ji#, **Yan Pan**#, Jiajun Lei# et al.
[10.1016/j.arr.2026.103124](https://doi.org/10.1016/j.arr.2026.103124) 

De Novo Design of Anti-Aging Peptides

Yan Pan[#], Li Fan[#], Nan Zhang[#] et al.

[10.1101/2025.10.04.680435](https://doi.org/10.1101/2025.10.04.680435) 

BioRxiv preprint

Oct 2025

Revealing 1, 3-diphenylpropane's coagulation toxicity via infomaxnet-based network toxicology and molecular simulations

Yan Pan[#], Hongxia Cai[#], Yufeng Ran et al.

[10.1016/j.j.ecoenv.2025.118834](https://doi.org/10.1016/j.j.ecoenv.2025.118834) 

Ecotoxicology and Environmental Safety

Aug 2025

ElixirSeeker: A Machine Learning Framework Utilizing Attention-Driven Fusion of Molecular Fingerprints for the Discovery of Anti-Aging Compounds

Yan Pan[#], Hongxia Cai[#], Fang Ye[#] et al.

[10.1101/2024.09.08.611839](https://doi.org/10.1101/2024.09.08.611839) 

Aging Cell

May 2025

WormCNN-Assisted Establishment and Analysis of Glycation Stress Models in *C. elegans*: Insights into Disease and Healthy Aging

Yan Pan, Zhihang Huang, Hongxia Cai et al.

[10.3390/ijms25179675](https://doi.org/10.3390/ijms25179675) 

International Journal of Molecular Sciences

Sep 2024

L-Theanine Extends the Lifespan of *Caenorhabditis elegans* by Reducing the End Products of Advanced Glycosylation

Zhihang Huang, Haiming Jing, **Yan Pan**, Hongxia Cai et al.

[10.3390/foods14020221](https://doi.org/10.3390/foods14020221) 

Foods

Jan 2025

The joint toxicity effect of glyphosate and cadmium in a concentration-dependent manner on nematode *Caenorhabditis elegans*

Zhihang Huang[#], Anastasia Ngozi Ezemaduka[#], Hongxia Cai[#], **Yan Pan**[#], Yiwen Gong et al.

[10.1016/j.j.ecoenv.2024.117081](https://doi.org/10.1016/j.j.ecoenv.2024.117081) 

Ecotoxicology and Environmental Safety

Oct 2024

Reduction of DNA Topoisomerase Top2 reprograms the epigenetic landscape and extends health and life span across species

Man Zhu, Meng Ma, Lunan Luo, Feiyang Li, Jiashun Zhang, **Yan Pan**, Lu Yang et al.

[10.1111/accel.70010](https://doi.org/10.1111/accel.70010) 

ELife/Aging Cell

Nov 2024

Optimization of polyethylene terephthalate biodegradation using a self-assembled multi-enzyme cascade strategy

Lizhu Aer, Qifa Jiang, Linling Zhong, Qiuyue Si, Xianghong Liu, **Yan Pan**, Juan Feng et al.

[10.1016/j.jhazmat.2024.134887](https://doi.org/10.1016/j.jhazmat.2024.134887) 

Journal of Hazardous Materials

Oct 2024

Circular RNA CDR1as/ciRS-7- a novel biomarker in solid tumors

Yun Zhang, Chanyu Xiong, Zhilin Jiang, Xiao Wang, Juanjuan Ji, **Yan Pan**, Tianshu Yu et al.

[10.3389/fonc.2024.1468363](https://doi.org/10.3389/fonc.2024.1468363) 

Frontiers in Oncology

Nov 2024

Conference Presentations

(Oral Presentation) **De novo design of Anti-Aging Peptides**

Oct 2025

IUTOX 17th International Congress of Toxicology

(Oral Presentation) **Deep Learning and Molecular Dynamics-Based Network Toxicology: Unraveling the molecular mechanism of styrene dimers affecting the coagulation system**

Oct 2024

The 7th Conference on Toxicity Testing Alternatives and Translational Toxicology

Honors and Awards

Silver Medal(Team Advisor) , 2024 International Genetically Engineered Machine (iGEM)	Nov 2025
Special Prize(Team Leader) , "Challenge Cup" National College Student Series Science and Technology Academic/Entrepreneurship Plan Competition of UESTC	Apr 2025
Gold Medal(Team Advisor) , 2024 International Genetically Engineered Machine (iGEM)	Oct 2024
Silver Medal(Team Leader) , the 11th National Undergraduate Medical Innovation Competition and the "Belt and Road" International Competition	Sep 2024
Young Outstanding Paper Award , Chinese Society of Toxicology	Aug 2024
Gold Medal(Team member) , 2023 International Genetically Engineered Machine (iGEM)	Oct 2023

About Me

My works focus on Using **machine learning** to computationally address **complex biological systems** such as aging.

My goal is to use **quantitative biology** to explore some of the deeper questions in science and philosophy. For instance, what is the ultimate mechanism of aging? Is the universe governed by strict laws, or is there an element of randomness—does “God roll the dice”? By applying rigorous, data-driven approaches to biological processes, I hope to gain insights that bridge the gap between scientific understanding and philosophical inquiry.